

Effects of maternal employment and sibling's gender on housework time allocation.

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Abstract

Millions of women in Latin America have entered the labor market during the past three decades, but housework and care are still primarily undertaken by women. While the international literature on housework time allocation is extensive, evidence for Latin America—and how domestic work is distributed among all household members, including children—is limited. This study analyzes the division of housework in Mexican households with two parents and two children aged 12 to 21, focusing on the children's gender and birth order. Using the 2014, 2019, and 2024 rounds of Mexico's National Time Use Survey (ENUT), we estimate a Seemingly Unrelated Regressions (SUR) model of each family member's housework time and share. We find that maternal employment leads to a redistribution of housework toward husbands and children, even though mothers remain the primary contributors. Daughters, especially those with a brother, take on more housework, and they compensate for this by working in paid jobs instead of studying. These gendered patterns are conditional relationships that remain constant across time, consistent with long-standing gender norms.

Keywords: Intrahousehold time allocation, family economics, unpaid work, sibling gender composition, gender norms, children

1. Introduction

Over the past three decades, women's labor force participation has moved toward parity with men's across the industrialized world—Goldin's (2014) "Grand Gender Convergence" (Olivetti & Petrongolo, 2016). The household sector has not kept up with the changes. According to Becker's (1965) household-production framework, extended by Gronau (1977), increased female participation adjusts one input without renegotiating the technology of domestic production, and unpaid-work gaps have decreased at roughly half the rate of market gaps (Pailhé et al., 2021). The same mechanism can be seen in Latin America, where female participation increased from 41.3% in 1990 to 53.3% in 2022 (Gontero & Vezza, 2023), but women still perform 70% of housework (Rubiano-Matulevich & Viollaz, 2019), one of the region's major disparities (Amarante & Rossel, 2018). This residual gap is our economic target. When a woman enters the labor market, the time she frees up at home is reallocated among the remaining family, subject to a constraint that the regional literature has yet to identify directly.

According to Sullivan et al. (2018), Mexico's uneven progress toward gender equality is hindered by institutional barriers, such as the gender division of unpaid care work and labor market segmentation. Female labor force participation increased, but this trend did not extend to unpaid work: even after the COVID-19 crisis, women still perform most of the domestic work in Mexican households (Hoehn-Velasco et al., 2022). The average household is a well-defined production unit—71% nuclear, 53.8% two-parent, according to the National Institute of Statistics and Geography (Instituto Nacional de Estadística y Geografía, INEGI), with 2.1 children per woman—so each of the four members plays a definite role (INEGI, 2017). Within-country heterogeneity is itself informative: Skoufias and Parker (2001) show that under the PROGRESA conditional cash-transfer program, income shocks caused girls to absorb domestic work disproportionately, while López-Chávez and Ortiz-Ávila (2024) demonstrate that dual-earner couples converge on childcare but not on total housework—evidence

that working mothers reallocate the residual within the household rather than renegotiating it with their partners. This makes Mexico a country that provides an illustrative context for analysis.

The literature on the “second shift” documents the burden of domestic work that falls on adults (Giménez-Nadal & Sevilla, 2012; Hochschild & Machung, 2012). This agenda has been developed for Latin America based on regional studies: Campaña et al. (2017) link gender norms to the distribution of total work in Mexico, Peru, and Ecuador; Amarante and Rossel (2018) and Amarante et al. (2024) document the trade-offs between income and domestic tasks in Uruguay and throughout Latin America and the Caribbean (LAC); Rubiano-Matulevich and Viollaz (2019) provide the regional benchmark on time allocation; García-Román and Martínez-Mendiola (2025) demonstrate that economies of scale occur only when households have four members. However, these studies either aggregate children’s contributions or do not consider sibling gender composition as a source of variance within countries. That is the gap we are aiming for.

We frame the analysis around three questions that jointly test substitution against the persistence of gender norms (Goldin, 2014; Sullivan et al., 2018). First, how does the maternal employment decision shift the joint distribution of housework across the four members? Adult-couple evidence predicts a moderate mother-to-father shift (Pailhé et al., 2021), while the children’s literature predicts that daughters bear more of the residual (Cordero-Coma & Esping-Andersen, 2018; Schulz, 2021). Second, does sibling gender composition modify that redistribution? Beckerian substitution implies gender-symmetric sibling contributions; if prevailing gender norms and task-allocation conventions dominate, daughters should absorb more when they co-reside with a brother. Because our data are repeated cross-sections, we cannot discriminate among internal socialization, a necessity-driven response to maternal time scarcity, and shared cultural norms; we therefore report conditional associations consistent with these mechanisms, not causal transmission (Blair, 1992; Coltrane, 2000; West & Zimmerman, 1987). Third, is the maternal reduction traded against children’s paid work and study time, and does it depend on birth order?

We use aggregated data from the 2014, 2019, and 2024 rounds of Mexico’s National Time Use Survey (Encuesta Nacional sobre el Uso del Tiempo, ENUT), which collects activity diaries from all household members aged 12 and older. We focus on households with two parents and two children aged 12 to 21, for a total sample of 3,467 households (921 in 2014, 1,346 in 2019, and 1,200 in 2024). Because the rounds were drawn under different sampling frames, the estimates are calculated using the unweighted pooled sample to preserve comparability. The design contributes in three respects: it identifies all four working-age members simultaneously through Seemingly Unrelated Regressions (SUR) equations (following Campaña et al., 2017); it uses sibling gender composition as within-Mexico variation; and it tests whether housework absorption is traded against children’s paid work and study time. Pooling the 2014, 2019, and 2024 rounds allows us to analyze the data—at the population level, across repeated cross-sections rather than within families—as a dimension that spans across all three questions rather than as a separate one.

We observe that mothers remain the primary contributors but that daughters absorb more housework—especially alongside a brother—and that this gendered pattern is stable across the three rounds. The work is structured as follows: Section 2 reviews the literature, Section 3 develops the theoretical framework, Section 4 describes the data, Section 5 the SUR methodology, Section 6 the results, and Section 7 concludes with policy implications and limitations.

2. Literature review

Several studies have examined the causes and effects of the gendered division of household work on families and society (Coltrane, 2000; Lachance-Grzela & Bouchard, 2010). Women continue to do most of the housework, particularly in households with children (Bianchi, 2000; Leopold et al., 2018; Sullivan et al., 2018). Gender differences in unpaid labor among adult couples have decreased, but at a rate that is about half that of market-work gaps. The gender-convergence literature shows that although women’s market labor has gotten closer to men’s, the domestic division of labor has changed

more slowly and has stagnated in some settings since the mid-2010s (Goldin, 2014; Leopold et al., 2018; Sullivan et al., 2018). The same asymmetry appears among children: across Denmark, Germany, Italy, and the United States, the gender gap is present early and widens with age, with daughters contributing substantially more than sons (Bonke, 2010; Dotti Sani, 2016; Schulz, 2021).

Two household characteristics condition children's contributions. Maternal employment raises children's participation, since time-constrained working mothers require more support at home; this effect is consistent for daughters but weaker and less stable for sons (Benin & Edwards, 1990; Blair, 1992; Gager et al., 1999). Sibling number and gender composition matter as well: Housework per child falls with the number of siblings, and the firstborn contributes more (Schulz, 2021; Solberg, 1994), while co-residing with a sibling reduces boys' housework but increases girls', so that families with daughters spend more time on housework overall (Bonke, 2010; Brody & Steelman, 1985).

2.1 Housework and gender inequality in Latin American countries

This evidence was generated almost entirely for high-income countries with long-standing time-use instruments. Since 2017, regional literature has begun to address the parallel gap for LAC, and recent stand-alone time-use surveys across the region—in countries such as Chile, Costa Rica, El Salvador, Guatemala, Uruguay, and Peru—make broader comparison more viable. Rubiano-Matulevich and Viollaz (2019), using time-use data from a group of LAC countries, including Mexico, Colombia, Ecuador, and Peru, provide a regional benchmark for market-household time allocation and show that women's share of unpaid work is high throughout the region, with Mexico standing among the highest. According to Campaña et al. (2017), a gender-norms index from the World Values Survey relates to the distribution of total employment in Mexico, Peru, and Ecuador, indicating higher equality in more egalitarian societies. Amarante and Rossel (2018), in a comparative study covering several Latin American countries (Colombia, Mexico, Peru and Uruguay), document the persistence of women's housework share despite falling fertility, and Amarante et al. (2024) extend this evidence

to the same set of countries plus Chile, interpreting the positive correlation between women's earnings and their housework share as a Beckerian substitution channel operating at the adult level.

Domínguez-Amorós et al. (2021) cross-country analysis of Latin American economies indicates gendered reproductive work and slowed convergence since 2015. This raises the question of whether the intra-household gender gap among children has changed over time. García-Román and Martínez-Mendiola (2025) found that once households have four people, the per capita burden of housework stabilizes. Additionally, women's time spent on housework is maintained when shared with other women rather than men. These studies indicate that a high and consistent percentage of unpaid labor performed by women is a regional pattern rather than a characteristic of Mexico.

The evidence has become more detailed within Mexico. According to Skoufias and Parker (2001), the children's gendered division of domestic labor remained largely intact despite the exogenous financial shock of the PROGRESA transfers, with girls doing more cleaning, cooking, and food preparation than boys. Casique (2008) documented that the gendered division of unpaid work narrows only marginally with women's earnings, and López-Chávez and Ortiz-Ávila (2024), using the 2019 round of ENUT, found that dual-earner couples converge on childcare but not on total housework—consistent with the "second shift" (Hochschild & Machung, 2012). Hoehn-Velasco et al. (2022) documented that the COVID-19 shock produced gender asymmetries in Mexican time use that persisted through the recovery.

Two features of this literature limit what can be inferred about intra-household mechanisms. First, most studies aggregate adult members into a single household total, so the relative contributions of mothers, fathers, and children cannot be separated (Amarante & Rossel, 2018; Casique, 2008; Giménez-Nadal, 2015). Second, the few studies that decompose by member either contrast adults only or pool children's contributions without conditioning on sibling gender and birth order (Hoehn-Velasco et al., 2022; López-Chávez & Ortiz-Ávila, 2024). As a result, the LAC housework gap is

well documented, but the within-household mechanism—who absorbs the time a working mother releases—has been identified only indirectly, through cross-country adult regressions or household-size economies of scale. They also don't exploit the ENUT's repeated rounds to find out if these patterns are evolving over time.

Treating children as productive actors rather than recipients of parental time is the premise that makes the intra-household channel identifiable: even pre-adolescents contribute measurable, gendered amounts of housework (Bonke, 2010; Larson & Verma, 1999; McHale et al., 1999), a pattern also documented for low- and middle-income countries (Bornstein et al., 2016). The empirical literature interprets these patterns through necessity, within-household socialization, and prevailing cultural gender role expectations channels (Blair, 1992; Bonke, 2010; Cordero-Coma & Esping-Andersen, 2018; Fuwa, 2004; West & Zimmerman, 1987). Because they generate observationally similar predictions in cross-sectional data, we report conditional associations rather than causal transmission; we develop the three interpretations, and the reasons they cannot be separated, in the next section.

Against this background, we contribute to the expanding LAC literature in three ways. First, we identify each of the four members—father, mother, older sibling, and younger sibling—simultaneously, estimating SUR equations for both minutes per day and each member's share of the household total; the LAC literature has applied this design to adults but not to children based on sibling position. Second, we use sibling gender composition (Boy/Boy, Boy/Girl, Girl/Boy, Girl/Girl, with birth order) as a within-Mexico variation to identify how the housework a mother releases is absorbed by the remaining members—a design used in Germany (Cordero-Coma & Esping-Andersen, 2018; Schulz, 2021) and Italy (Dotti Sani, 2016) but, to our knowledge, novel for Latin America. Third, we use the same framework to test whether children's housework is traded against their paid work and study time, a within-household test of the time-availability perspective. Cutting across these three contributions, we pool the 2014, 2019 and 2024 ENUT rounds to assess whether

the gendered patterns we document persist over a decade of rising female labor-force participation—an intertemporal question the region’s largely single-round studies have not addressed.

3. Theoretical background

The household combines market goods with members’ time to produce domestic commodities (Becker, 1965). Gronau’s (1977) extension partitions time into market work, household work and leisure, with the optimal allocation equating each member’s marginal productivities across uses. When a mother enters the labor market, her marginal home productivity is unchanged, but her opportunity cost of domestic time rises, so the household must reallocate the released housework across the remaining members under a binding total-time constraint. This reallocation—rather than any renegotiation of the production technology itself—is the object we estimate, and the total-time constraint is precisely the cross-equation dependence our SUR specification captures.

Three complementary perspectives structure the gendered division of routine family work (Coltrane, 2000; Lachance-Grzela & Bouchard, 2010) and generate different sign predictions. Under time availability, partners divide housework according to time spent outside the home (Davis et al., 2007; Hiller, 1984), so maternal employment shifts the load onto the father and children in a gender-neutral way. Under relative resources, external resources such as income and education confer power to negotiate (Lundberg & Pollak, 1996; Mannino & Deutsch, 2007), so an earning mother reduces her load to the extent her negotiating position allows—consistent with Beckerian specialization (Becker, 1981). Under gender ideology, housework confirms gender identities and operates largely independently of productivities or labor-market arrangements (Berk, 1985; West & Zimmerman, 1987), predicting that female members absorb more housework even where the gender-neutral lenses predict symmetry (Kühhirt, 2012; Leopold et al., 2018).

The contrast is testable. Under the first two lenses, siblings' contributions to the mother's released housework should be gender-symmetric; under the third, daughters should absorb disproportionately more, particularly when co-residing with a brother. Sibling gender composition is the variation that discriminates between these predictions.

The empirical literature offers three interpretations of children's gendered housework, and a central concern of our design is that they are observationally equivalent in cross-sectional data. The necessity channel holds that children substitute for time-constrained parents (Blair, 1992; Bonke, 2010). The internal-socialization channel holds that children internalize gendered patterns by observing and imitating parents in a shared household (Cordero-Coma & Esping-Andersen, 2018; Hu, 2015). The cultural norms channel attributes and the same patterns to prevailing societal expectations—correlated with context such as rurality and education—that are exogenous to the individual household (Campaña et al., 2018; Fuwa, 2004; West & Zimmerman, 1987).

All three predict the same signs: daughters spend more time on housework than sons, the gap widens in households with a brother, and a working mother's released time is absorbed primarily by female members. Because the mechanisms coincide in sign, the estimated coefficients cannot identify which one operates; separating them would require longitudinal. We therefore interpret our estimates as conditional associations consistent with these mechanisms operating jointly, not as evidence adjudicating between them.

Pooling three survey rounds spanning a decade adds a temporal dimension to this framework. The gender-convergence literature holds that, as female labor-force participation rises and gender attitudes modernize, the domestic division of labor should gradually become more egalitarian (Goldin, 2014; Sullivan et al., 2018). If this dynamic reached the intra-household allocation among children, the sibling gender-composition effect should attenuate across rounds. If prevailing gender norms remain deeply entrenched, the effect should persist despite rising maternal employment. Because our data

are repeated cross-sections of different households rather than a panel, this comparison describes the evolution of a population-level association across cohorts of households; it does not track norms as they are transmitted within families over time.

The framework yields three hypotheses, each carrying an explicit sign prediction the SUR estimates can confront. Under the gender-neutral logic shared by the time-availability and relative-resources lenses, maternal employment should reduce the mother's housework time while raising that of the father and both children (H1). The gender-ideology lens departs from this gender-neutral benchmark in a specific, testable way: daughters should absorb more of the mother's released housework than sons, and a daughter's contribution should be larger when she co-resides with a brother than with a sister (H2)—a pattern gender-neutral substitution cannot generate. Extending time availability to the children's own margin, those who absorb more housework should reduce their paid work or study time (H3). Finally, given the decade spanned by our pooled sample, the gender-convergence and entrenched-norms views make opposing predictions about whether the composition effects in H2 attenuate or persist across rounds; we assess these effects through the $\text{year} \times \text{composition}$ interactions.

4. Data

To analyze family members' participation in housework, we pool three rounds of the ENUT: 2014, 2019 and 2024 (INEGI, 2025). The ENUT is a repeated cross-sectional survey designed to measure paid and unpaid work and how men and women allocate time to domestic production. Time-use surveys are commonly used to analyze individual time-allocation decisions (Amarante et al., 2024; Bianchi, 2000; Campaña et al., 2020; Casique, 2008; Folbre, 1994; Folbre et al., 2005; Giménez-Nadal & Sevilla, 2012).

Because the three rounds were drawn under different sampling frames—the 2024 round transitioned to INEGI's updated Master Sampling Frame—their expansion factors are not directly comparable and cannot be combined in a pooled sample. We therefore estimate all models and report all

descriptive statistics on the unweighted sample, which preserves comparability across rounds. The three rounds are conceptually comparable in the dimensions relevant to our analysis: they share the same household definition, the same reference population (individuals aged 12 and older), and a common activity classification, the Mexican Classification of Time-Use Activities (Clasificación Mexicana de Actividades de Uso del Tiempo, CMAUT), so that our measure of housework is constructed across rounds.

Each round records the hours and minutes each household member aged 12 and older devotes to housework, with two reference periods: workweek and weekend. Our target sample consists of households formed by a father, a mother, and two children aged 12 to 21 living at home; parents must be of different genders and under or equal to 60 years old.¹ These households may also include other residents, such as children under 12 or people with mobility disabilities, who do not participate in household chores but whose presence we consider as controls. This selection provides a sample of rigorously reduced heterogeneity, allowing us to identify each position within the household—father, mother, older and younger sibling—and its respective contribution to housework. The final pooled sample comprises 3,467 households: 921 in 2014, 1,346 in 2019 and 1,200 in 2024.

The time each member spends on housework is obtained by summing the minutes devoted to each housework task during the workweek and dividing by five to obtain an average per day. Housework comprises cooking, general household cleaning (washing dishes, sweeping, taking out the garbage), caring for plants or pets, washing and ironing clothes, maintaining household vehicles, purchasing food, goods and services for the household, and paying and administering household bills.²

¹ We use the full 12–21 age range because narrowing it (for example, to those under 18) would further reduce a sample of four participants that is already selective, and, given that the probability of continuing to live with parents varies by gender at older ages, it could in itself introduce a gender-based selection bias regarding who remains under observation; furthermore, in Mexico, young people tend to leave their parents' home relatively late (García-Andrés et al., 2021), so the 12- to 21-year-old range covers most adolescents who live with their parents.

² See List A1 for complete list of questions.

Table 1 Descriptive household statistics by ENUT survey round, 2014–2024

	2014	2019	2024	Total	χ^2 / t-test
N	921 (26.7%)	1,346 (38.8%)	1,200 (34.6%)	3,467 (100%)	
Total household time (min per day)	9hrs 22min	10hrs 3min	9hrs 41min	9hrs 39min	<0.001
Maternal employment status	52.4%	57.8%	56.8%	56%	0.031
Sibling gender composition					
Boy/Boy	27.2%	27.3%	27%	27.2%	0.977
Boy/Girl	24.3%	25.4%	24.8%	24.9%	
Girl/Boy	25.4%	25%	24.4%	24.9%	
Girl/Girl	23%	22.3%	23.8%	23%	
Older sibling's age	17.2	17.4	17.4	17.4	0.116
Age difference (older-younger sibling)	3	3.2	3.2	3.2	0.120
Households with ≥ 1 girl under 12	30.7%	29.5%	27.6%	28.8%	0.196
Households with ≥ 1 boy under 12	31.1%	27.4%	25.1%	27.6%	0.001
Households with ≥ 1 child under 6	19.5%	16%	16.5%	17.1%	0.017
Mother's age	40.8	40.9	41.2	41	0.160
Older sibling employment status	32.8%	40.2%	39.1%	37.8%	<0.001
Mother's education					
Elementary	36.5%	24.1%	24.3%	27.5%	<0.001
Middle school	34%	41.7%	37.7%	38.3%	
High school	16.5%	17.7%	20.9%	18.5%	
College	13%	16.5%	17.1%	15.7%	
Person with a mobility impairment	7.2%	6.1%	6.6%	6.6%	0.537
Domestic employee	3.2%	3.2%	3.6%	3.4%	0.885
Rural area	26.7%	26.7%	27.7%	26.3%	0.094

Notes: Continuous variables report means; categorical variables and under-12 indicators report the percentage of households. Differences across survey rounds were tested using one-way ANOVA (continuous) and Pearson's χ^2 (categorical); p-values are in the final column. Given the change in INEGI's sampling frame in 2024, statistics are reported without expansion factors to preserve comparability across rounds. Source: Own estimates using data from ENUT 2014, 2019 and 2024.

Table 1 presents descriptive statistics by survey round. The estimation sample remains steady during the decade, with no significant difference in sibling gender composition across rounds (χ^2 p=0.977), as are the older sibling's age (p=0.116), the sibling age gap (p=0.120) and the mother's age (p=0.160). This stability supports pooling the three rounds and anticipates the temporal persistence we document in the results. What does change over time is consistent with broader trends: maternal labor force

participation rises across rounds, from 52.4% in 2014 to 57.8% in 2019 and 56.8% in 2024 ($p=0.031$); maternal education improves, with the share of mothers with at most elementary schooling falling from 36.5% in 2014 to 24.3% in 2024 ($p<0.001$); and the presence of young children declines. Total household housework time also differs significantly across rounds ($p<0.001$), reflecting level shifts that our year controls absorb.

Table 2: Descriptive household statistics by maternal employment status, pooled sample 2014–2024

	Maternal employment status			χ^2 / t-test
	Non-working	Working	Total	
N	1,524 (44%)	1,943 (56%)	3,467 (100%)	
Total household time (min per day)	9hrs 49min	9hrs 30min	9hrs 39min	0.024
Sibling gender composition				
Boy/Boy	27%	27.3%	27.2%	0.244
Boy/Girl	25.6%	24.4%	24.9%	
Girl/Boy	23.4%	26%	24.8%	
Girl/Girl	24%	22.3%	23.1%	
Older sibling's age	17.2	17.5	17.4	<0.001
Age difference (older-younger sibling)	3.2	3.2	3.2	0.067
Households with ≥ 1 girl under 12	32.6%	25.8%	28.8%	<0.001
Households with ≥ 1 boy under 12	31.8%	24.4%	25.6%	0.001
Households with ≥ 1 child under 6	21.1%	13.9%	17%	<0.001
Mother's age	40.8	41.2	41	0.046
Older sibling employment status	36.8%	38.6%	37.8%	0.299
Mother's education				
Elementary	37%	19.9%	27.5%	<0.001
Middle school	38.9%	37.8%	38.3%	
High school	15.8%	20.6%	18.5%	
College	8.3%	23.7%	15.7%	
Person with a mobility impairment	7.7%	5.6%	6.6%	0.014
Domestic employee	1.9%	4.5%	3.4%	<0.001
Rural area	35.1%	19.3%	26.3%	<0.001

Notes: Estimation sample of four-member households, split by maternal employment status. Continuous variables report means; categorical variables and under-12 indicators report the percentage of households. Differences between working and non-working mothers tested using mean-comparison tests (continuous) and Pearson's χ^2 (categorical); p-values in the final column. Reported without expansion factors. Source: Own estimates using data from ENUT 2014, 2019 and 2024.

Table 2 reports the same statistics by maternal employment status over the pooled sample. Working mothers account for 56% of households, reflecting the specific population studied—relatively young mothers, averaging 41 years of age, with two adolescent children. Crucially for our identification, the two subsamples are balanced on the dimensions central to the analysis: sibling gender composition does not differ significantly between households with working and non-working mothers (χ^2 $p=0.244$). The subsamples do differ, as expected, on characteristics reflecting the positive selection of working mothers into education and urban residence—maternal schooling ($p<0.001$), rural residence (35.1% vs. 19.3%, $p<0.001$), the number of young children ($p<0.001$), and paid domestic help ($p<0.001$). Because all of these characteristics enter the SUR specification as controls, the estimated associations are conditioned on them. The older sibling is employed in 37.8% of households. Of households, 17% include at least one child under 6, 6.6% include a person with a mobility impairment, and 3.4% hire paid domestic help.

5. Methodology

The empirical literature has commonly used independent OLS regressions to estimate the time each family member spends on housework (Amarante & Rossel, 2021; Casique, 2008; Leopold et al., 2018; Schulz, 2021). However, the time one family member devotes to a household task is time another member does not, so estimating each member’s housework time independently ignores the mechanical interdependence among them. To account for this joint structure, we estimate a SUR model, which considers the housework time of all family members simultaneously and allows their error terms to be correlated. Prior work has used SUR to analyze intra-household time allocation (Campaña et al., 2018, 2020).³

³ We observe that SUR increases efficiency by taking advantage of the cross-equation correlation of the errors, but it cannot resolve endogeneity or simultaneity in the regressors on its own. As a result, our estimates describe conditional associations rather than causal effects, a point we revisit in the Limitations section.

We pool three rounds of the ENUT—2014, 2019 and 2024—into a single repeated cross-section. For a given household i , let TF_i , TM_i , TO_i , TY_i denote the average time in minutes per day spent on housework during the workweek by the father, mother, older sibling and younger sibling, respectively.

The system is estimated as:

$$TF_i = \alpha_{tf} + \beta_1^{TF} Mat_emp_i + \beta_2^{TF} Comp_i + \beta_3^{TF} Year_i + \beta_4^{TF} (Comp_i \times Year_i) + \beta_5^{TF} X_i + \varepsilon_{tfi} \quad (1)$$

$$TM_i = \alpha_{tm} + \beta_1^{TM} Mat_emp_i + \beta_2^{TM} Comp_i + \beta_3^{TM} Year_i + \beta_4^{TM} (Comp_i \times Year_i) + \beta_5^{TM} X_i + \varepsilon_{tmi} \quad (2)$$

$$TO_i = \alpha_{to} + \beta_1^{TO} Mat_emp_i + \beta_2^{TO} Comp_i + \beta_3^{TO} Year_i + \beta_4^{TO} (Comp_i \times Year_i) + \beta_5^{TO} X_i + \varepsilon_{toi} \quad (3)$$

$$TY_i = \alpha_{ty} + \beta_1^{TY} Mat_emp_i + \beta_2^{TY} Comp_i + \beta_3^{TY} Year_i + \beta_4^{TY} (Comp_i \times Year_i) + \beta_5^{TY} X_i + \varepsilon_{tyi} \quad (4)$$

Mat_emp_i equals 1 if the mother is in the labor force and 0 otherwise; a mother is considered employed if she worked at least one hour during the reference week and received payment.

The vector $Comp_i$ comprises the four gender compositions a two-child household can have—Boy/Boy, Boy/Girl, Girl/Boy and Girl/Girl—where the first term denotes the older sibling's gender and Boy/Boy is the reference category (Schulz, 2021). $Year_i$ is a set of survey-round indicators (2014, 2019 and 2024), with 2024 as the reference year; the interaction $Comp_i \times Year_i$ allows the effect of sibling composition to differ across rounds, so that its coefficients test whether the gendered pattern of housework changed over the decade. The vector X_i collects household controls: the number of girls under 12, the number of boys under 12, and the number of children under 6 (each entered as a continuous count); age of the member of the family and the age gap between siblings; the older sibling's employment status; maternal education; total household housework time; the presence of a person with a disability; whether the household hires paid domestic help and rural residence. The terms ε_{tfi} , ε_{tmi} , ε_{toi} , ε_{tyi} are jointly distributed error terms.

The SUR framework allows these errors to be freely correlated across equations, capturing the constraint that the total time a household allocates to housework is finite: time one member spends is

time another does not. Including the counts of younger children under 12 and under 6 addresses the possibility that housework attributed to the focal siblings partly reflects the care of these younger children in the household, thereby allowing the identification of the sibling-composition effect without the influence of such caregiving demands.

We estimate the same specification using each member's share of total household housework as the dependent variable. Let $ShareF_i$, $ShareM_i$, $ShareO_i$, $ShareY_i$ denote each member's share, defined as:

$$Share_{ji} = \frac{\text{Time of the member } j}{\text{Total time of the household } i} \times 100 \quad (5)$$

where $j = \text{father, mother, older, younger}$ (Craig & Powell, 2018; Craig et al., 2015; Schulz, 2021).

6. Results

6.1 Housework time allocation

Table 3 reports the SUR estimates for time spent on housework (left) and share of total housework (right) for each family member, with the older sibling appearing first in the sibling gender composition. Coefficients on sibling composition are measured at the reference year (2024); the year and year $\times$ composition terms capture temporal variation, discussed below.⁴

When the mother works, all other members increase their housework time (and share) while the mother reduces hers. The mother reduces her housework by 68.81 minutes per day, or 11.77 percentage points (pp) of the household total. Correspondingly, the father increases his by 28.15 minutes (4.88 pp), the older sibling by 22.24 minutes (3.87 pp), and the younger sibling by 18.41

⁴ Only selected variables are presented; see Table A1 for full results.

minutes (3.02 pp). This joint reallocation—the mother reducing while every other member increases—is consistent with H1, the gender-neutral prediction shared by the time-availability and relative-resources perspectives.

Sibling gender composition shapes how much children contribute. Relative to a younger son in a two-son household, a younger daughter with an older brother spends 29.32 more minutes per day (5.67 pp) on housework. Relative to an older son in a two-son household, an older daughter with a younger brother spends 39.85 more minutes (8.25 pp). In two-daughter households, the older daughter spends 35.95 more minutes (6.75 pp) and the younger 28.69 more minutes (5.25 pp) than their counterparts in two-son households. When daughters have a brother, they do more housework than when they have a sister. This asymmetry is consistent with H2: Daughters absorb more housework and do so more strongly in the presence of a brother—a pattern that gender-neutral substitution cannot provide.

The parents' allocation also responds to the presence of daughters. Whenever there is a daughter, the mother reduces her housework load; the father also reduces his in the compositions Boy/Girl, especially when both children are girls. These patterns align with the time-availability perspective and the necessity channel of children's participation under parental time constraints (Blair, 1992; Bonke, 2010; Cordero-Coma & Esping-Andersen, 2018).

This within-household pattern echoes the regional evidence of García-Román and Martínez-Mendiola (2025): per-capita housework plateaus in four-member households because women share the residual with other women, not with men. Our SUR estimates identify directly which female member absorbs the marginal minute. However, these patterns are also consistent with prevalent cultural gender norms—the same expectations about women's household roles that influence both adults' and children's behavior, regardless of any active within-family transmission. Bornstein et al. (2016) document the same aggregate pattern across low- and middle-income countries: girls work more inside the household, boys outside. Our estimations show that this pattern operates through intra-

household absorption within the average four-member Mexican household, but the cross-sectional structure of the data does not allow us to identify which channel is at work.

Table 3: Seemingly Unrelated Regression estimates of housework time and shares, pooled sample 2014–2024

	Time				Share			
	Father	Mother	Older sibling	Younger sibling	Father	Mother	Older sibling	Younger sibling
Mother works								
Yes	28.15*** (2.51)	-68.81*** (3.94)	22.24*** (2.74)	18.41*** (2.44)	4.88*** (0.41)	-11.77*** (0.66)	3.87*** (0.44)	3.02*** (0.38)
Sibling gender composition (2024 ref.)								
Boy/Boy	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Boy/Girl	-9.44* (5.69)	-17.42* (8.93)	-2.45 (6.21)	29.32*** (5.54)	-1.47 (0.93)	-3.91*** (1.49)	-0.29 (0.99)	5.67*** (0.87)
Girl/Boy	-8.23 (5.73)	-25.43*** (8.99)	39.85*** (6.25)	-6.21 (5.58)	-0.95 (0.94)	-6.67*** (1.50)	8.25*** (1.00)	-0.64 (0.88)
Girl/Girl	-18.42*** (5.75)	-46.23*** (9.02)	35.95*** (6.27)	28.69*** (5.60)	-2.65*** (0.94)	-9.35*** (1.51)	6.75*** (1.00)	5.25*** (0.88)
Year								
2014	-13.51** (5.93)	28.18*** (9.30)	-11.00* (6.47)	-3.68 (5.77)	-1.67* (0.97)	3.47** (1.56)	-1.73* (1.03)	-0.08 (0.91)
2019	4.42 (5.36)	15.44* (8.41)	-7.75 (5.85)	-12.11** (5.22)	0.51 (0.88)	1.69 (1.41)	-0.78 (0.93)	-1.43* (0.82)
Year×composition (join test)								
p-value	0.417	0.542	0.859	0.744	0.313	0.324	0.443	0.28
Controls ^a	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	-17.32*** (6.57)	79.59*** (10.09)	-1.43 (10.44)	-61.80*** (9.96)	8.37*** (1.08)	72.29*** (1.68)	15.90*** (1.64)	3.27** (1.55)
Observations	3,444	3,444	3,444	3,444	3,444	3,444	3,444	3,444
R-squared	0.21	0.57	0.31	0.30	0.10	0.14	0.12	0.11

Notes: Time is the number of minutes per day dedicated to housework; share is the percentage of total household housework performed by each member. The "Year × composition (joint test)" row reports the p-value of the null that all year × composition interactions equal zero within each equation. ^aControls include number of girls under 12, number of boys under 12, children under 6, age of the member of the family and age gap, older sibling's employment status, maternal education, total household housework time, the presence of a person with a disability, domestic employee, and rural residence. Standard errors in parentheses. *** p<0.01, ** p<0.05 and * p<0.1. Reported without expansion factors. Source: Own estimates using data from ENUT 2014, 2019 and 2024.

The year's main effects suggest level shifts, while the year × composition interactions determine if the composition impact itself changed. A combined test cannot reject the null hypothesis that all such

interactions are zero in any equation, for time or shares (joint-test p-values range from 0.28 to 0.86).⁵ The gendered allocation of housework among siblings remains steady from 2014 to 2024. We interpret this as population-level persistence over multiple cross-sections; it describes the persistence of a contemporaneous associate rather than evidence of intergenerational transmission.

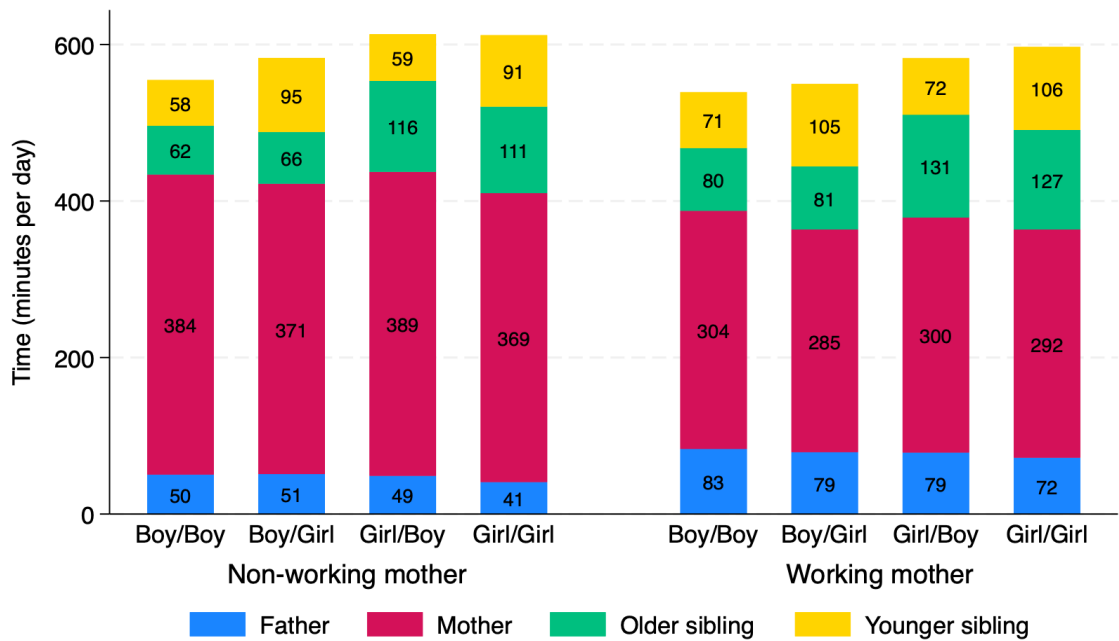


Fig. 1 Predicted time spent on housework on weekdays for all household members, by mother's employment status, pooled sample 2014–2024

⁵ The full set of year $\times$ composition interaction coefficients, together with the joint-test p-values for each equation, is reported in Table A3.

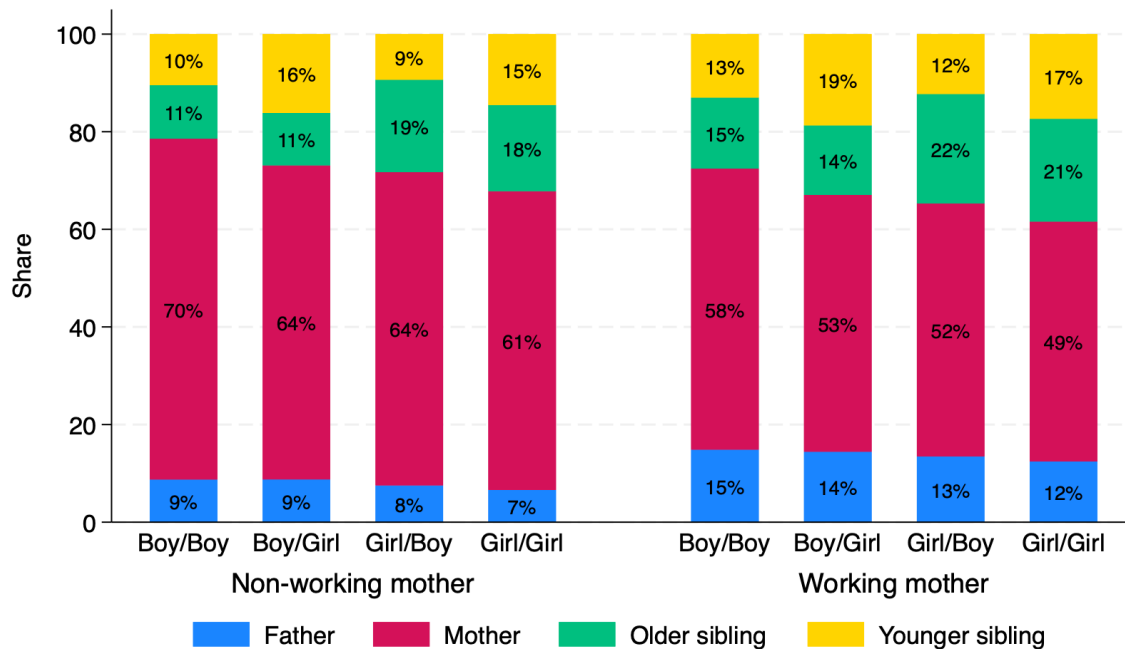


Fig. 2 Predicted shares of total housework on weekdays for all household members, by mother's employment status, pooled sample 2014–2024

Using the estimated coefficients, Figures 1 and 2 present the predicted times and shares for each member across the four household types, by maternal employment status (older sibling first on the x-axis).⁶ As expected, the mother spends the most time on housework regardless of her employment status. Where mothers do not work, they average roughly 370 to 390 minutes per day (Fig. 1), or 61% to 70% of the household total (Fig. 2); where they work, between 285 and 305 minutes (Fig. 1), or 49% to 58% (Fig. 2). Working mothers consequently devote less time to housework, both in absolute and relative terms.

Fathers spend less time on housework than mothers, but when the mother works, they increase their contribution—from roughly 41 to 51 minutes in non-working-mother households to 72 to 83 minutes when the mother works (Fig. 1), or from 7–9% to 12–15% of the household total (Fig. 2). Fathers'

⁶ Predicted values from the pooled SUR model of housework time and shares (Table 3), 2014–2024 ENUT rounds, averaged across rounds. Without expansion factors.

participation is therefore dependent on maternal employment, as suggested by the time-availability and relative-resource perspectives (Amarante et al., 2024; Bittman et al., 2003).

Fathers' participation is lower in households with a daughter, especially two daughters, regardless of maternal employment. In two-son households, the father's contribution is similar to the older son's. Households with two daughters spend more time on housework overall, and households with two sons less, which is consistent with prior literature. Where siblings are of different genders, daughters devote more time to housework regardless of birth order or maternal employment. These results are consistent with the persistence of a strong gender ideology, in which traditional gender attitudes contribute to an unequal division of housework among family members.

6.2 Leisure, paid work, and study time allocation

Because time is a zero-sum constraint, the housework that daughters disproportionately absorb must be drawn from their other activities. We therefore examine, within the same framework, how time devoted to paid work, study, and leisure responds to maternal employment and sibling gender composition, providing a within-household test of H3—that children who take on more housework reduce their market or study time.

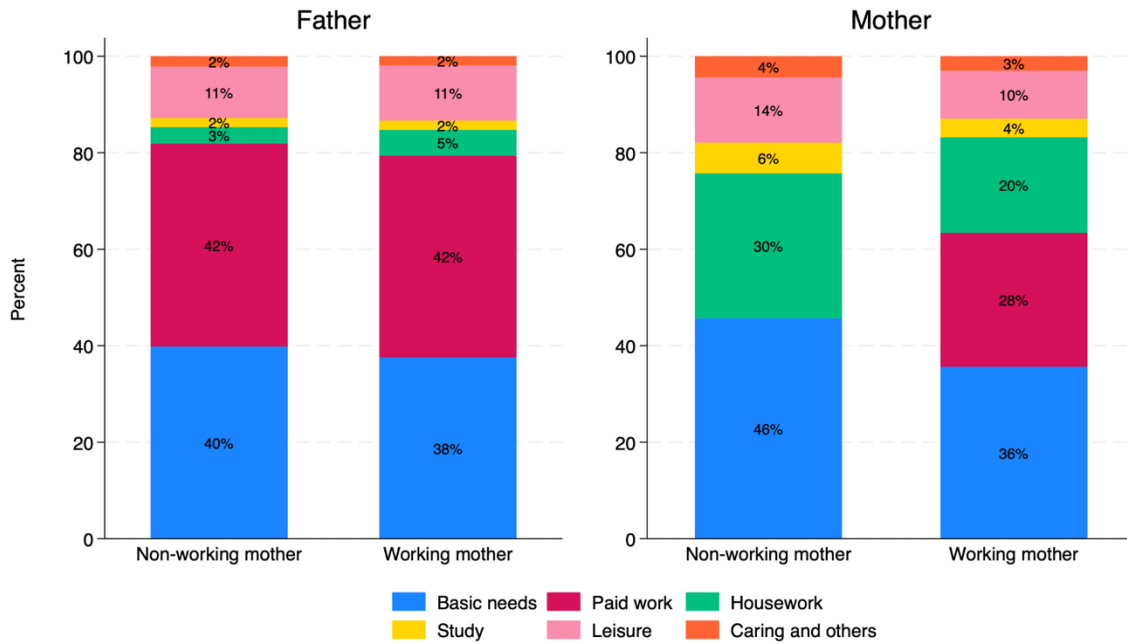


Fig. 3 Percentage of total time on weekdays of fathers and mothers dedicated to each activity, by mother's employment status, pooled sample 2014–2024

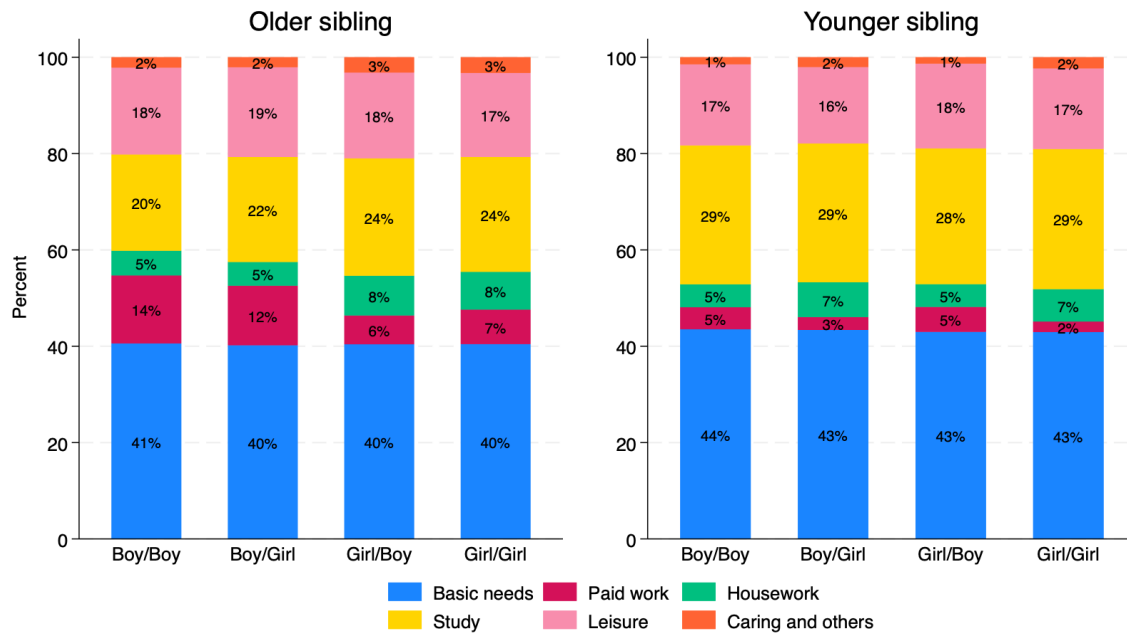


Fig. 4 Percentage of total time on weekdays of older and younger siblings dedicated to each activity, by sibling composition, pooled sample 2014–2024

Figures 3 and 4 report the average shares of time family members devote to basic needs, paid work, housework, study, leisure, and care.⁷ As Figure 3 shows, the father's time allocation is generally unchanged by the mother's employment, whereas working mothers considerably reduce the shares devoted to all other activities, including basic needs. Figure 4 shows that, regardless of the younger sibling's gender, an older son works more in the market, while an older daughter does more housework—though she also studies more. Among younger siblings, a younger daughter does more housework without studying less.

Table 4 Seemingly Unrelated Regression estimates of paid work and study time, pooled sample 2014–2024

	Paid work time				Study time	
	Father	Mother	Older sibling	Younger sibling	Older sibling	Younger sibling
Mother works						
Yes	5.52 (7.55)	392.79*** (5.53)	-14.76*** (5.40)	6.52 (4.82)	24.23*** (8.19)	13.71** (6.62)
Sibling gender composition (2024 ref.)						
Boy/Boy	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Boy/Girl	14.09 (17.12)	2.93 (12.54)	-9.92 (12.25)	-33.96*** (10.94)	17.13 (18.58)	22.16 (15.03)
Girl/Boy	14.09 (17.24)	-3.72 (12.62)	-37.66*** (12.33)	32.12*** (11.01)	22.38 (18.70)	-30.59** (15.13)
Girl/Girl	-8.63 (17.30)	5.46 (12.67)	-31.26** (12.38)	-15.64 (11.05)	8.91 (18.78)	13.47 (15.19)
Year						
2014	49.89*** (17.83)	3.28 (13.05)	-5.25 (12.75)	-4.58 (11.39)	-1.08 (19.35)	3.81 (15.65)
2019	40.75** (16.13)	-16.78 (11.81)	0.14 (11.54)	-1.32 (10.30)	50.64*** (17.50)	40.40*** (14.15)
Year×composition (join test)						
p-value	0.410	0.173	0.987	0.628	0.802	0.423
Controls ^a	Yes	Yes	Yes	Yes	Yes	Yes
Constant	642.16*** (33.43)	95.58*** (26.20)	-183.06*** (26.11)	-239.76*** (23.31)	530.53*** (39.62)	550.37*** (32.04)
Observations	3,444	3,444	3,444	3,444	3,444	3,444
R-squared	0.04	0.64	0.60	0.18	0.27	0.13

Notes: The time refers to the number of minutes per day dedicated to paid work and study. ^aControls include number of girls under 12, number of boys under 12, children under 6, age of the member of the family and age gap, older sibling's employment status, maternal education, total household

⁷ Observed average time shares computed directly from the pooled sample (2014–2024 ENUT rounds). Without expansion factors.

housework time, the presence of a person with a disability, domestic employee, and rural residence. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$ and * $p < 0.1$. Reported without expansion factors. Source: Own estimates using data from ENUT 2014, 2019 and 2024.

Table 4 reports SUR estimates for paid work time and study time.⁸ The main findings are: (a) when the mother works, neither the father nor the older child reduces paid work, while the older child studies marginally more; (b) daughters do less paid work when they are the older sibling (−37.66 minutes for Girl/Boy, −31.26 for Girl/Girl); and (c) the younger son studies less, but only when the older sibling is a girl. Notably, the reduction in the older daughter’s market work is not mirrored by a reduction in her study time—the housework she absorbs is traded against paid work rather than study time, consistent with H3. As with housework, the year $\times$ composition interactions are jointly non-significant for both paid work and study (joint-test p-values from 0.17 to 0.99; Table A3), indicating that these patterns also persist across the 2014–2024 period.⁹

7. Conclusion

Despite rising female labor participation, the division of housework in Mexico remains unequal, adding a "second shift" of unpaid work for women (Hochschild & Machung, 2012). Treating the family as a production unit with four members and pooling three ENUT rounds over a decade, our SUR estimates document three conditional associations. First, maternal employment is associated with a redistribution of housework away from the mother and toward the father and both children, though mothers remain the primary contributors—consistent with the gender-neutral prediction of the time-availability and relative-resources perspectives (H1). Second, daughters absorb more of the released housework and do so more strongly when they co-reside with a brother than with a sister, regardless of birth order—a pattern that gender-neutral substitution alone cannot generate (H2), exploring the relationship between sibling arrangement and gendered household behavior (Blair,

⁸ Only selected variables are presented; see Table A2 for full results.

⁹ A SUR for leisure time showed no statistically significant main coefficients.

1992; Goldschneider & Waite, 1991). Third, this absorption is traded against daughters' paid work time rather than their study time (H3), locating the adjustment on the labor market margin.

The gendered pattern has remained steady over the decade, with no significant year $\times$ composition interactions in housework, paid job, or study. As female labor-force participation increased between 2014 and 2024, households' allocation of domestic work among sons and daughters remained stable. This stability is consistent with the entrenched norms view rather than gender convergence at the intra-household level, and it supports the characterization of Mexico's gender revolution as uneven rather than stalled (Sullivan et al., 2018): the market margin has shifted while the domestic one, among children, has not. That finding captures population-level persistence over many cross-sections of different households.

These trends correspond to observationally identical mechanisms—necessity, within-household socialization, and prevalent cultural gender norms—which cross-sectional data cannot distinguish. Two aspects of our design, however, limit the interpretation. The sibling-composition effect is estimated based on rurality, maternal education, and the number and gender of household children under 12—the demographic margins where gender-role attitudes are most rigid in Mexico (Campaña et al., 2017; Casique, 2008) and caregiving demands are most likely to be misclassified as housework. Cultural standards appear to be at work, but they fail to account for a significant portion of the intra-household disparity.

Beyond the Mexican case, the finding carries a broader economic implication. The housework released by a working mother is absorbed within the household along gendered lines and drawn from daughters' market time—a channel through which the gender gap is reproduced in the next generation before those daughters ever enter the labor market. Identifying this channel at the household level, rather than through cross-country adult regressions or household-size economies of scale, is the contribution that aggregate LAC evidence (Amarante et al., 2024; Rubiano-Matulevich & Viollaz,

2019) cannot provide. The fact that the channel shows no signs of fading despite a decade of increased female employment highlights its policy relevance.

7.1 Policy implications

Because the redistributive burden of maternal employment falls on daughters—and shows no sign of attenuating over time—policy should act at both the household and the contextual level. Educational and media campaigns that challenge traditional gender stereotypes may weaken these associations from an early age, particularly in the rural, lower-education settings where norms are most rigid. Complementarily, workplace policies that raise the involvement of fathers and sons—paid paternity leave and gender-neutral parental incentives—target the margin our estimates identify as most unequal. That daughters' absorption is traded against their paid work rather than their time study is a partial reassurance for human-capital accumulation, but the early entrenchment of a domestic-work gap remains a concern for their later labor-market trajectories.

7.2 Limitations

Our estimates are conditional associations, not causal effects. SUR improves efficiency by allowing correlated errors across equations but does not address endogeneity or simultaneity in the regressors. Separating the mechanisms of necessity, socialization, and cultural norms—which all have the same sign—would require longitudinal or quasi-experimental variation.

The pooled data comprise independent samples of different households in each round, not the same households followed over time. Our temporal analysis therefore describes the evolution of a population-level association across cohorts of households; it cannot trace how housework behavior is transmitted within families or internalized by individuals as they form their households. This is why we frame the temporal finding as persistence of a population pattern rather than as evidence for or against intergenerational transmission.

Confining the analysis to two-parent households with two children aged 12 to 21 provides clean identification of each household position but excludes single-parent, extended, childless, and larger households, which may allocate housework differently. Because such households may also overrepresent settings where traditional norms operate most strongly—non-working mothers in our sample are more concentrated in rural areas and lower education—exercise caution when extrapolating the results to more egalitarian settings.

Although we control for the number and gender of children under 12 and for the presence of children under 6, the survey does not fully separate supervisory or emotional caregiving from the tasks we measure. The 12-to-21 age range for the focal siblings also spans distinct legal, educational, and labor transitions, so the estimates are averages that may not be uniform across a 12- and a 21-year-old; interactions by age bracket are a natural extension. Time use is self-reported and averaged over weekdays, smoothing within-week and weekend variation, and recall or social-desirability bias may affect reporting, particularly among adolescents.

Our analysis is limited to Mexico. Although recent stand-alone time-use surveys in several Latin American countries would possibly allow for a comparative analysis, the heterogeneity of these instruments and the change in INEGI's sampling frame in the 2024 round complicate even within-Mexico weighting across rounds. We thus contextualize our findings within the regional evidence at the descriptive and contextual levels, leaving a harmonized cross-country estimation, which would improve relevance, for future work. Additionally, more varied home configurations, longitudinal designs that follow allocation around maternal employment status, and paired activity data that tracks whether housework is performed alone or jointly (e.g., mother-daughter) could all be explored in future research.

Availability of data and material

The data used in the article are generated by INEGI, which stands for Instituto Nacional de Estadística y Geografía (National Institute of Statistics and Geography) which are freely available at <https://www.inegi.org.mx/programas/enut/2014/>, <https://www.inegi.org.mx/programas/enut/2019/> and, <https://www.inegi.org.mx/programas/enut/2024/#microdatos>.

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